

Setup

```
In [1]: import json
import numpy as np
import matplotlib.pyplot as plt
```

Description of the Visualization Function

The `analyze_frequencies` function performs comparative stylometric analysis between human-written and AI-generated Arabic text. The function:

- 1 . Data Loading: Loads frequency counts from JSON files containing POS (Part-of-Speech) or NER (Named Entity Recognition) tags for both human and AI text
- 2 . Data Cleaning: Removes unwanted/irrelevant tags like "UN_TAGGED", "S", "E", "", "O"
- 3 . Normalization: Calculates percentages based on total token counts for fair comparison
- 4 . Top-N Selection: Identifies the most frequent elements from human text to focus analysis
- 5 . Visualization: Creates horizontal bar charts comparing human vs AI usage percentages with value labels
- 6 . Analysis: The notebook applies this function across multiple scenarios:
 - Different AI models (ALLaM, Llama, Jais, OpenAI)
 - Different datasets (Social Media, Academic Abstracts)
 - Different generation methods (polishing, from title, from title+content)
 - Both POS and NER tag analysis

The function reveals stylometric differences between human and AI-generated Arabic text, helping identify patterns that could distinguish between the two sources.

```
In [2]: def analyze_frequencies(
    human_frequency_jsonpath,
    ai_frequency_jsonpath,
    top_n=10,
    title="Frequency Comparison",
    return_values=False,
):
    # Load data from JSON files
    with open(human_frequency_jsonpath, "r", encoding="utf-8") as f:
        human_freq = json.load(f)
    with open(ai_frequency_jsonpath, "r", encoding="utf-8") as f:
        ai_freq = json.load(f)

    # Remove unwanted labels
    exclude_labels = {"UN_TAGGED", "S", "E", "", "O"}
    human_freq = {k: v for k, v in human_freq.items() if k not in exclude_labels}
    ai_freq = {k: v for k, v in ai_freq.items() if k not in exclude_labels}

    # Get totals for normalization
    human_total = sum(human_freq.values())
    ai_total = sum(ai_freq.values())
```

```

# Get top N elements from human text
top_elements = sorted(human_freq.items(), key=lambda x: x[1], reverse=True)

# Extract data for plotting
labels = [elem[0] for elem in top_elements]
human_pct = [(human_freq[label] / human_total) * 100 for label in labels]
ai_pct = [(ai_freq.get(label, 0) / ai_total) * 100 for label in labels]

# Create horizontal bar chart
y_pos = np.arange(len(labels))
fig, ax = plt.subplots(figsize=(10, max(6, len(labels) * 0.4)))

bars1 = ax.barh(y_pos - 0.2, human_pct, 0.4, label="Human", alpha=0.8)
bars2 = ax.barh(y_pos + 0.2, ai_pct, 0.4, label="AI", alpha=0.8)

# Add value labels at end of bars
for i, (h_val, a_val) in enumerate(zip(human_pct, ai_pct)):
    ax.text(h_val + 0.1, i - 0.2, f"{h_val:.1f}%", va="center", fontstyle="italic")
    ax.text(a_val + 0.1, i + 0.2, f"{a_val:.1f}%", va="center", fontstyle="italic")

ax.set_yticks(y_pos)
ax.set_yticklabels(labels)
ax.set_xlabel("Percentage of Total Tokens")
ax.set_title(title)
ax.legend()
ax.invert_yaxis()

plt.tight_layout()
plt.show()

# Print summary stats
# print(f"Analyzed top {len(labels)} elements from human text")
# print(
#     f"Largest difference: {max(abs(h - a) for h, a in zip(human_pct, ai_pct))}"
# )

# Return data for further analysis
if return_values:
    return {
        "labels": labels,
        "human_pct": human_pct,
        "ai_pct": ai_pct,
        "ratios": [
            ai / human if human > 0 else 0 for ai, human in zip(ai_pct, human_pct)
        ],
    }

```

Social Media dataset

ALLaM

By Polishing

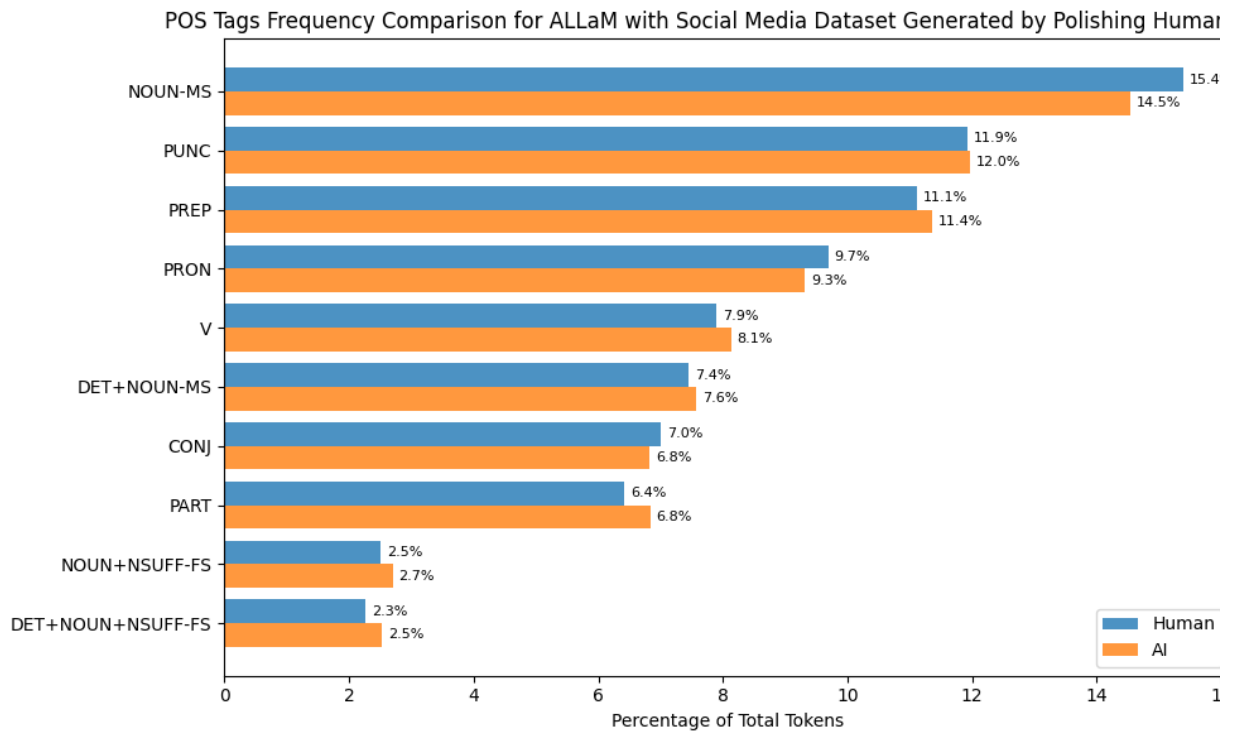
POS

```

In [3]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/social_media_human_freq.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da_ai_freq.json",
    title="POS Tags Frequency Comparison for ALLaM with Social Media Data"
)

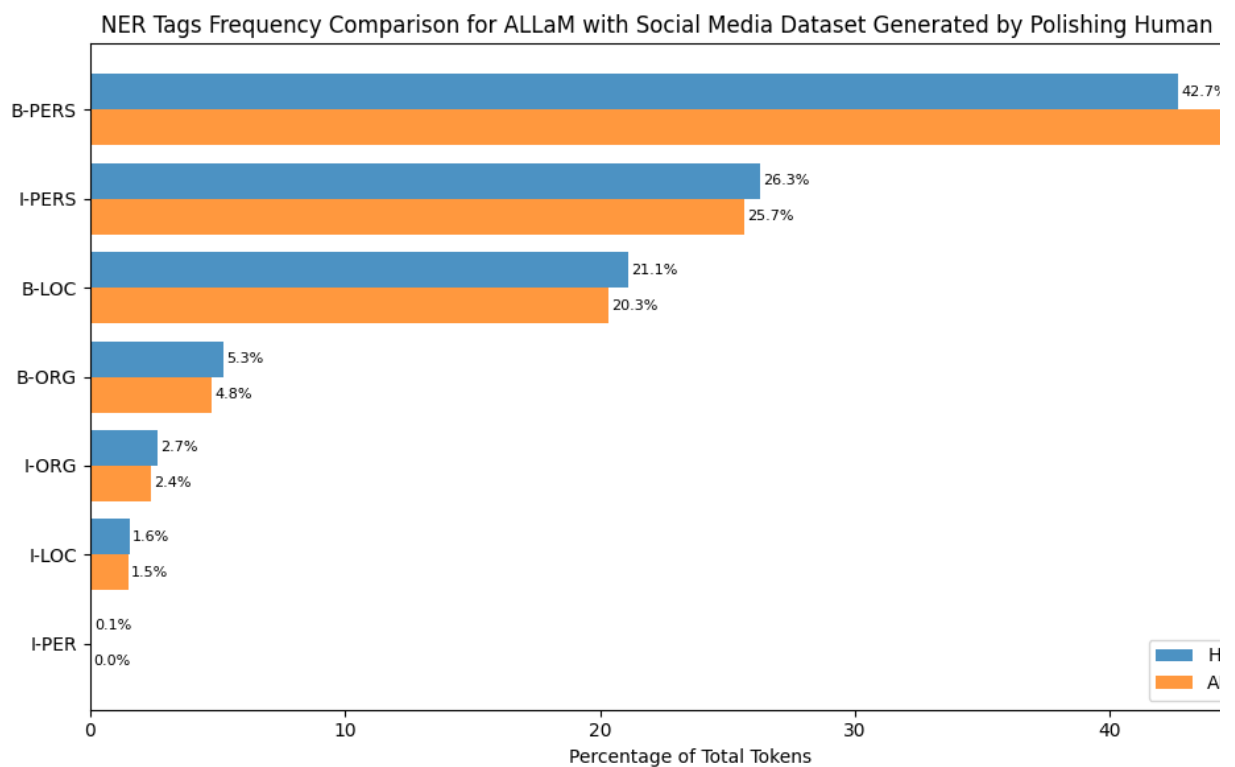
```

```
top_n=10,  
)
```



NER

```
In [4]: analyze_frequencies(  
    human_frequency_jsonpath="notebooks/stylometric_analysis/social_media  
    ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da  
    title="NER Tags Frequency Comparison for ALLaM with Social Media Data  
    top_n=7,  
)
```

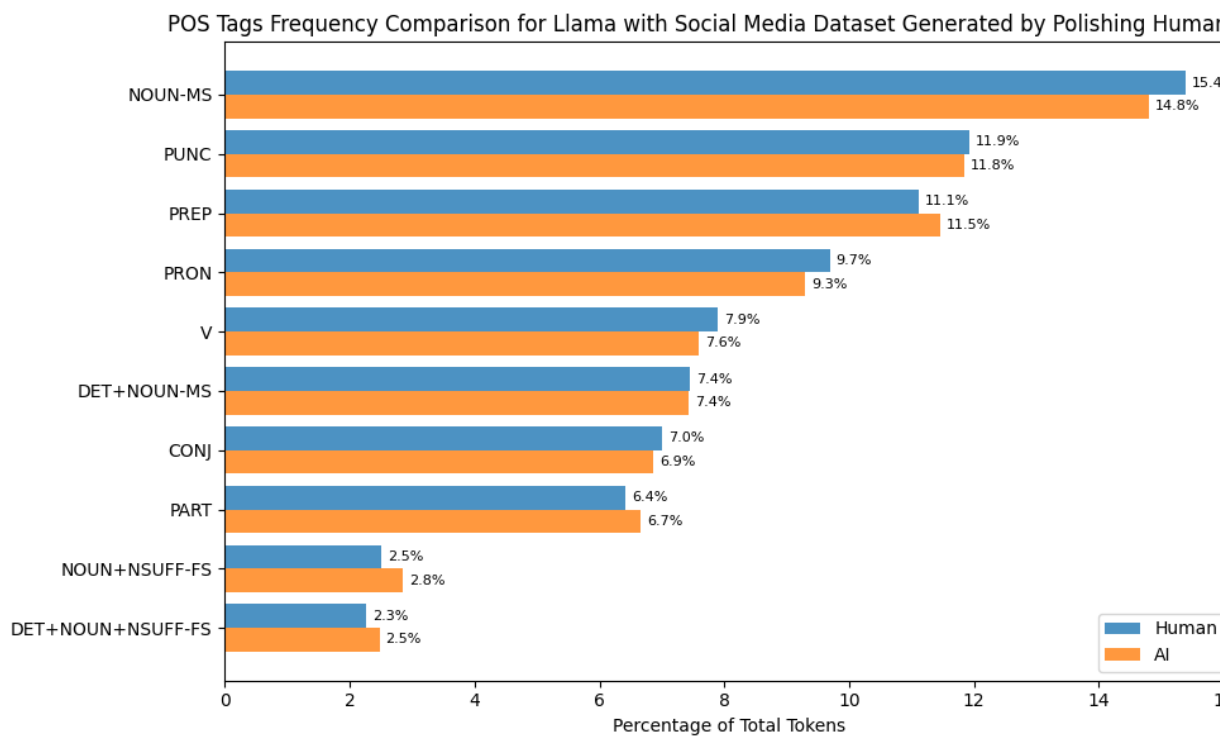


Llama

By Polishing

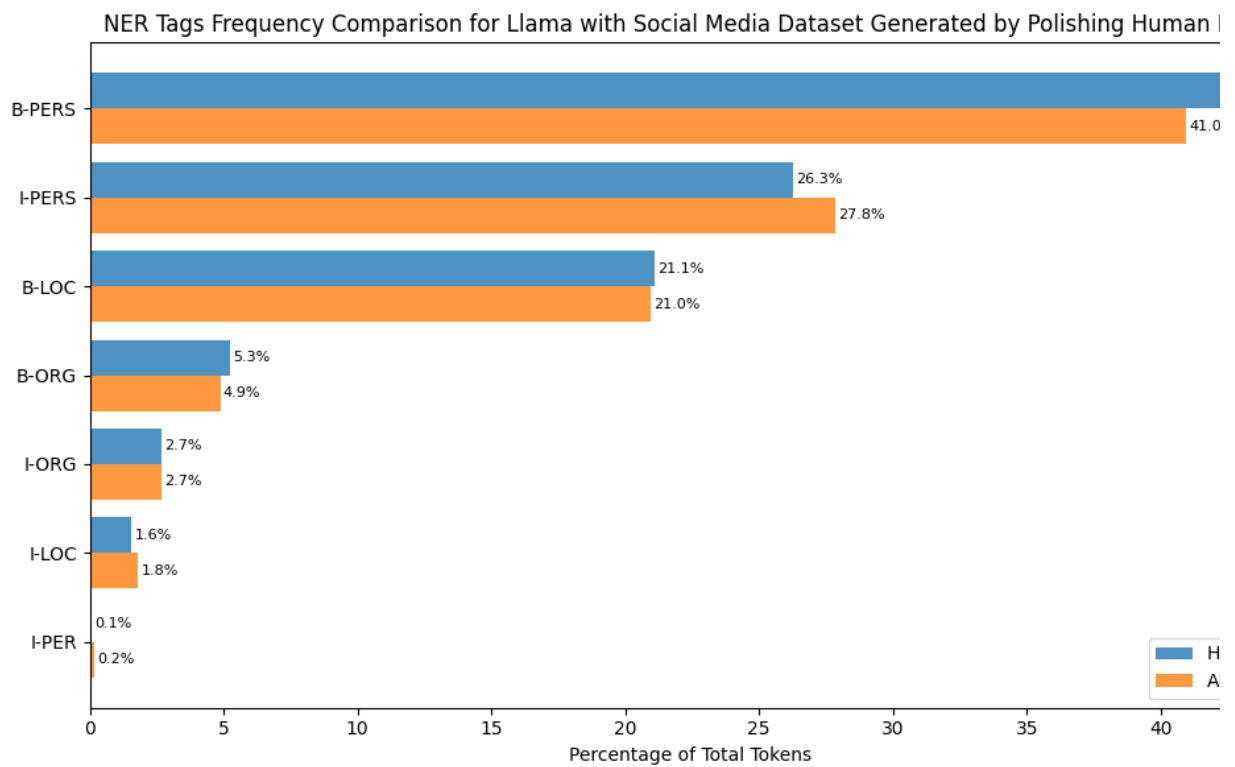
POS

```
In [5]: analyze_frequencies(  
    human_frequency_jsonpath="notebooks/stylometric_analysis/social_media  
    ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da  
    title="POS Tags Frequency Comparison for Llama with Social Media Data  
    top_n=10,  
)
```



NER

```
In [6]: analyze_frequencies(  
    human_frequency_jsonpath="notebooks/stylometric_analysis/social_media  
    ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da  
    title="NER Tags Frequency Comparison for Llama with Social Media Data  
    top_n=7,  
)
```

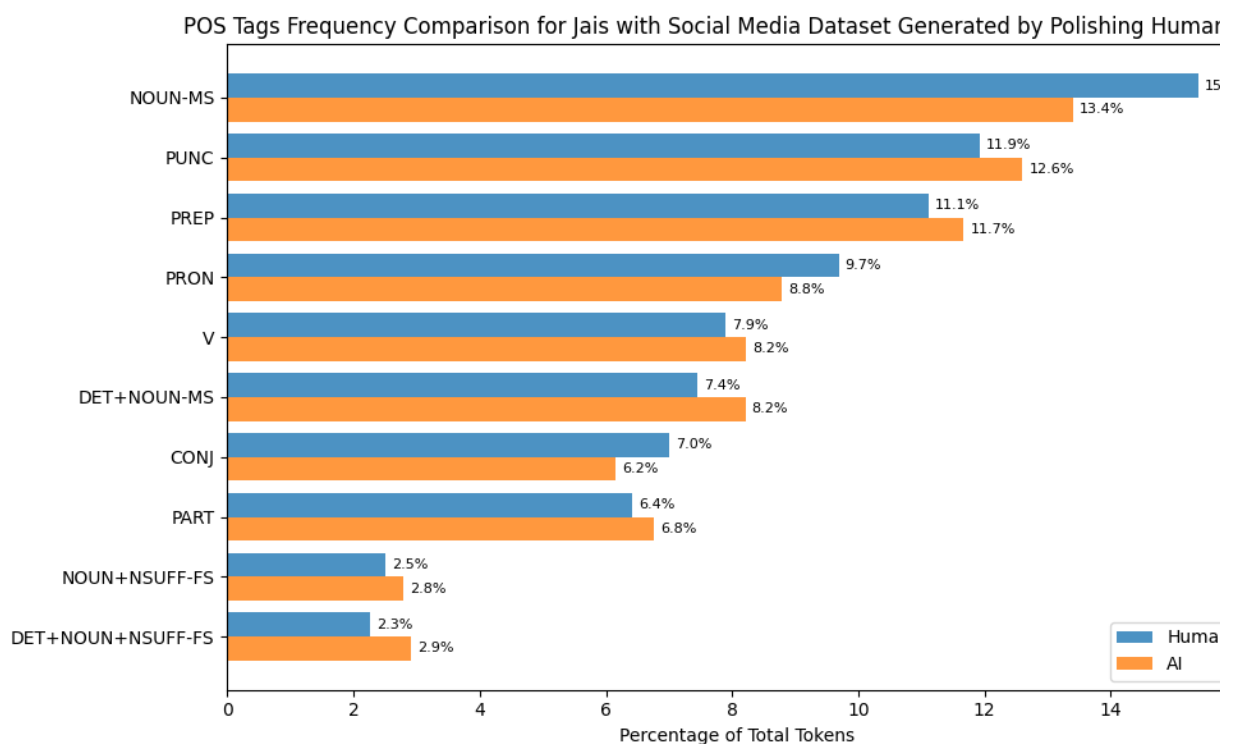


Jais

By Polishing

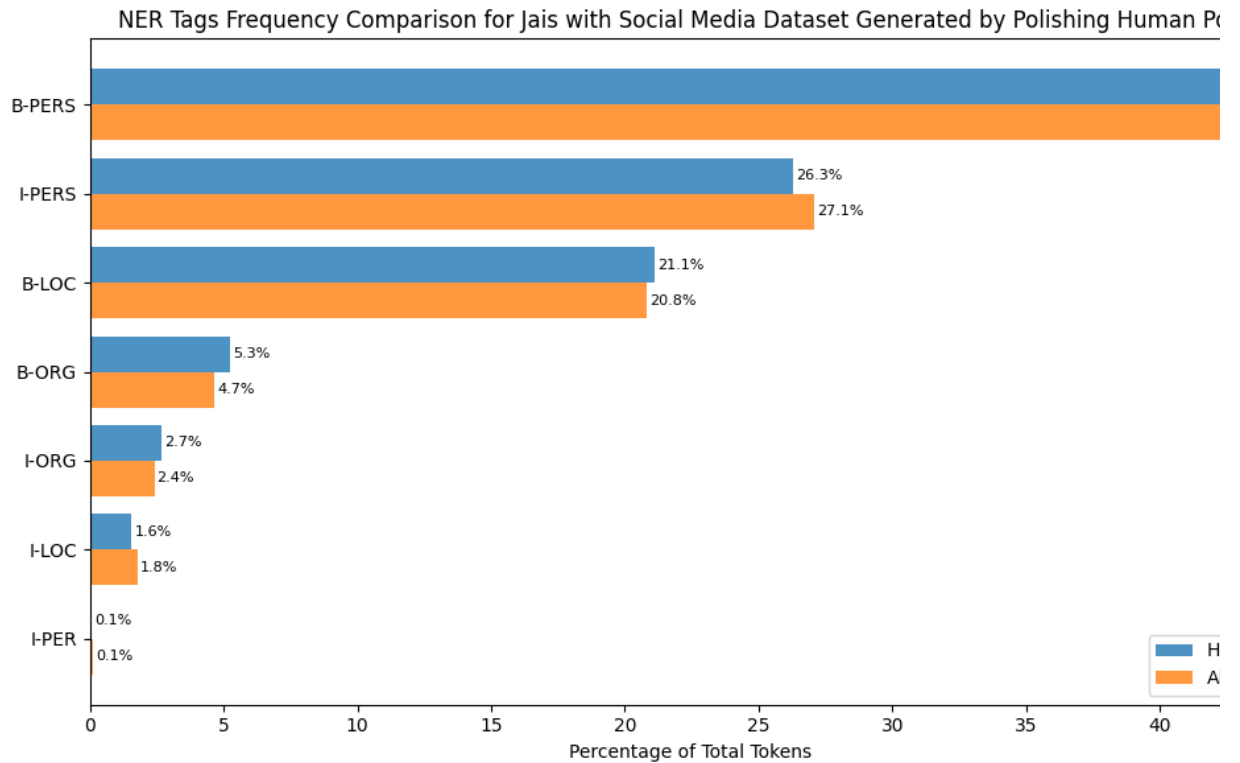
POS

```
In [7]: analyze_frequencies(
        human_frequency_jsonpath="notebooks/stylometric_analysis/social_media
        ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da
        title="POS Tags Frequency Comparison for Jais with Social Media Datas
        top_n=10,
    )
```



NER

```
In [8]: analyze_frequencies(  
    human_frequency_jsonpath="notebooks/stylometric_analysis/social_media  
    ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da  
    title="NER Tags Frequency Comparison for Jais with Social Media Datas  
    top_n=7,  
    )
```

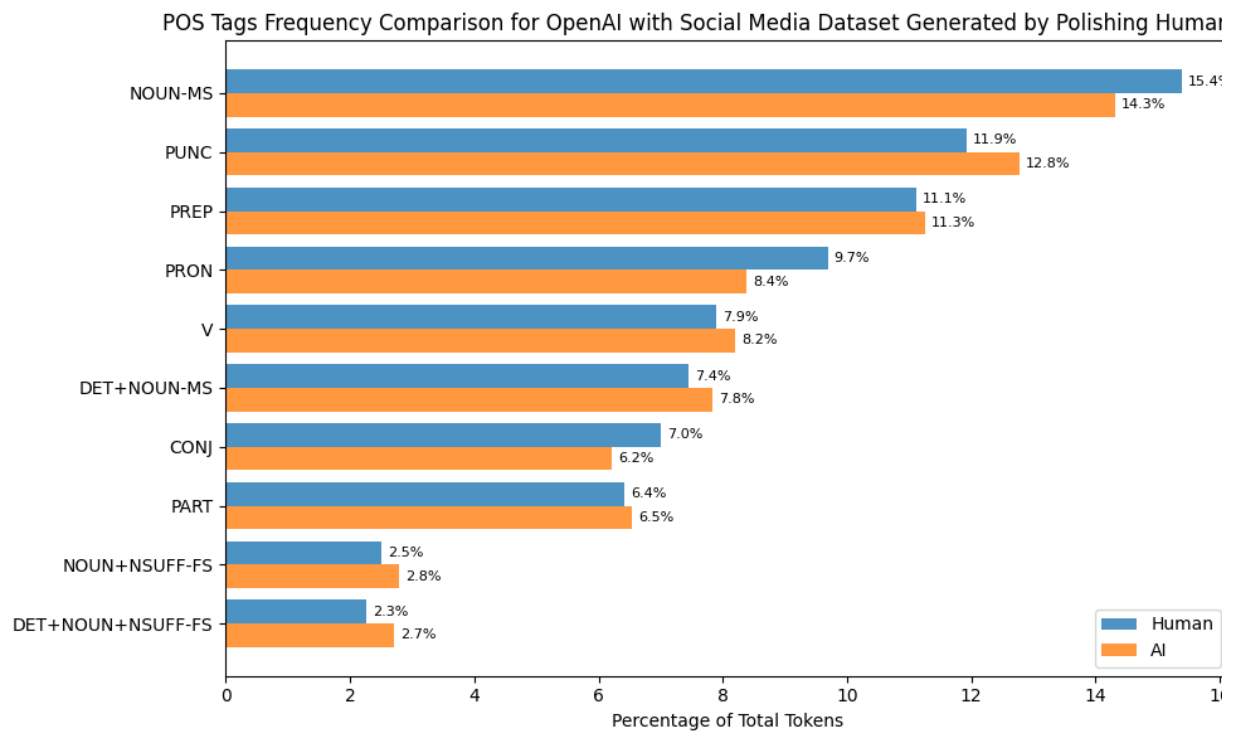


OpenAI

By Polishing

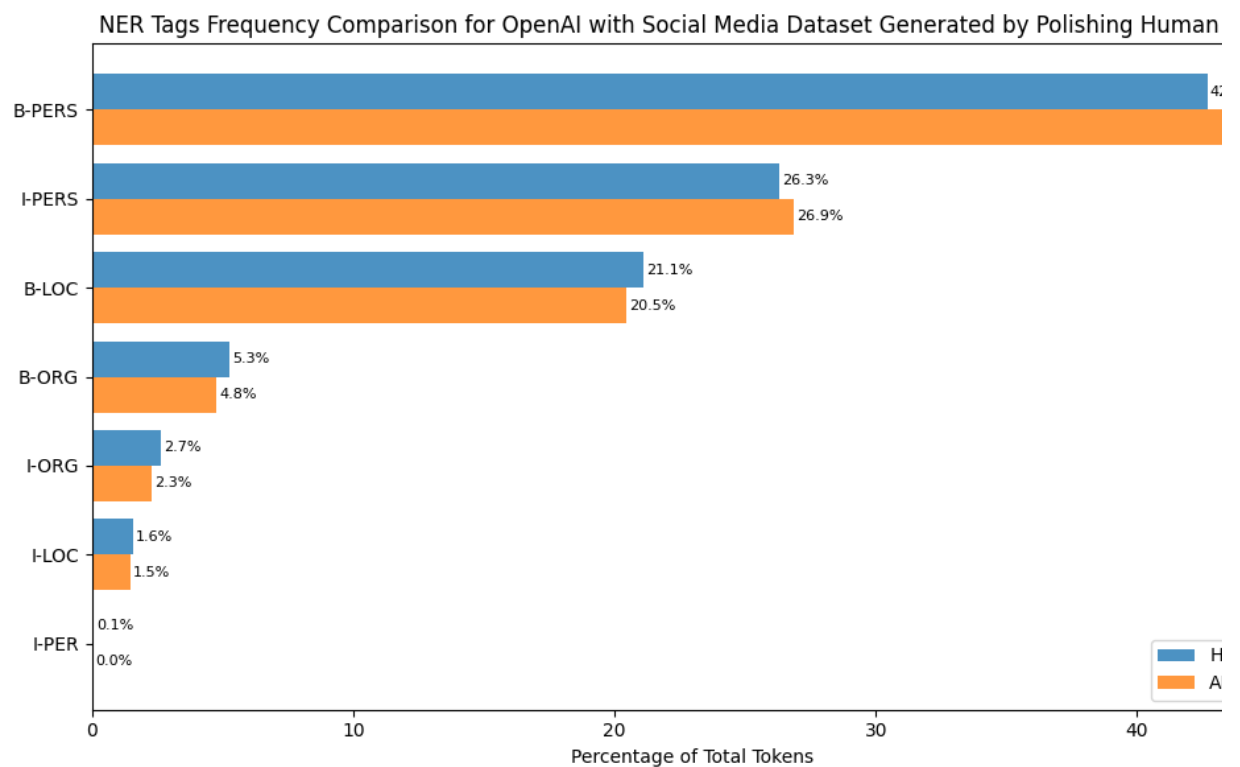
POS

```
In [9]: analyze_frequencies(  
    human_frequency_jsonpath="notebooks/stylometric_analysis/social_media  
    ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da  
    title="POS Tags Frequency Comparison for OpenAI with Social Media Dat  
    top_n=10,  
    )
```



NER

```
In [10]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/social_media_
    ai_frequency_jsonpath="notebooks/stylometric_analysis/social_media_da
    title="NER Tags Frequency Comparison for OpenAI with Social Media Dat
    top_n=7,
)
```



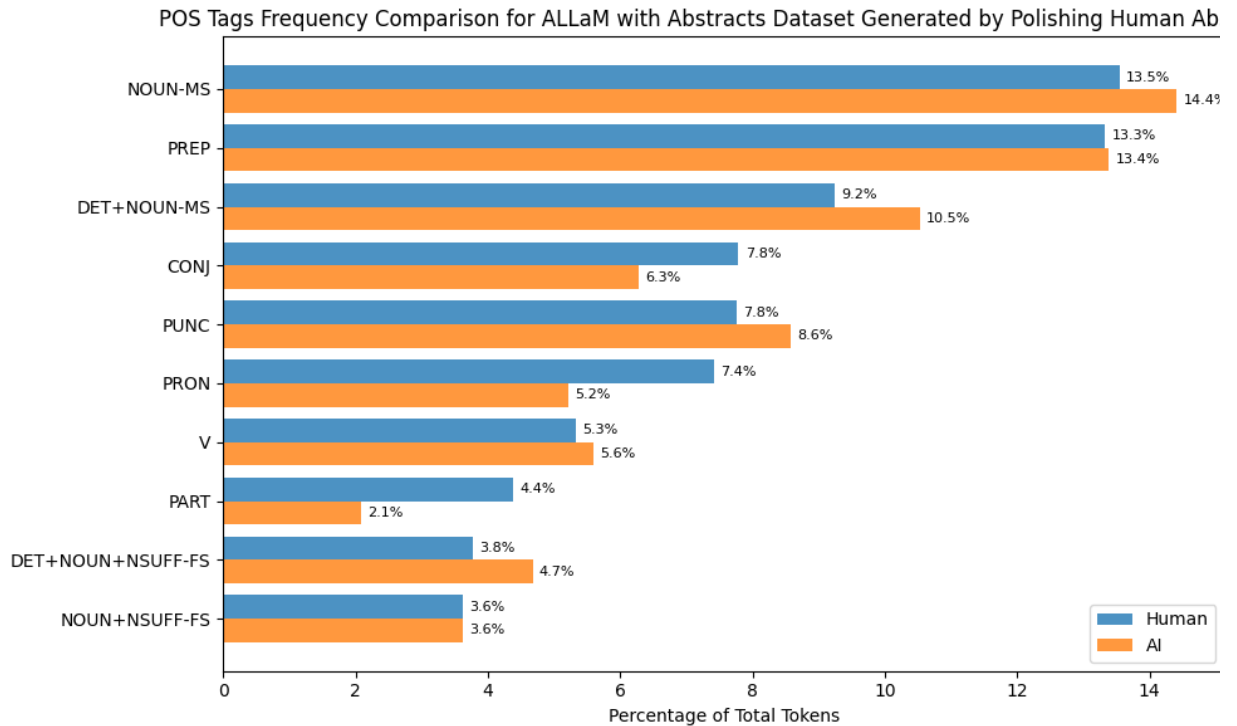
Abstracts Dataset

ALLaM

By Polishing

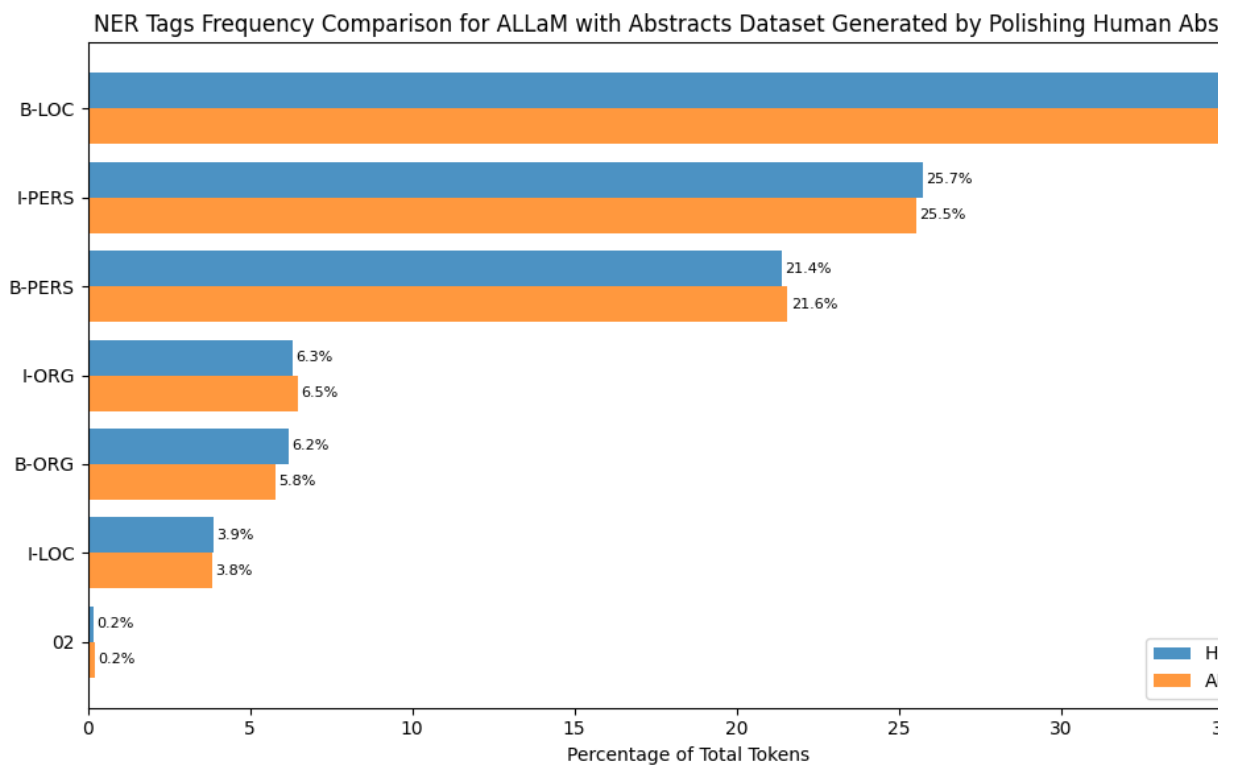
POS

```
In [11]: analyze_frequencies(  
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",  
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",  
    title="POS Tags Frequency Comparison for ALLaM with Abstracts Dataset",  
    top_n=10,  
)
```



NER

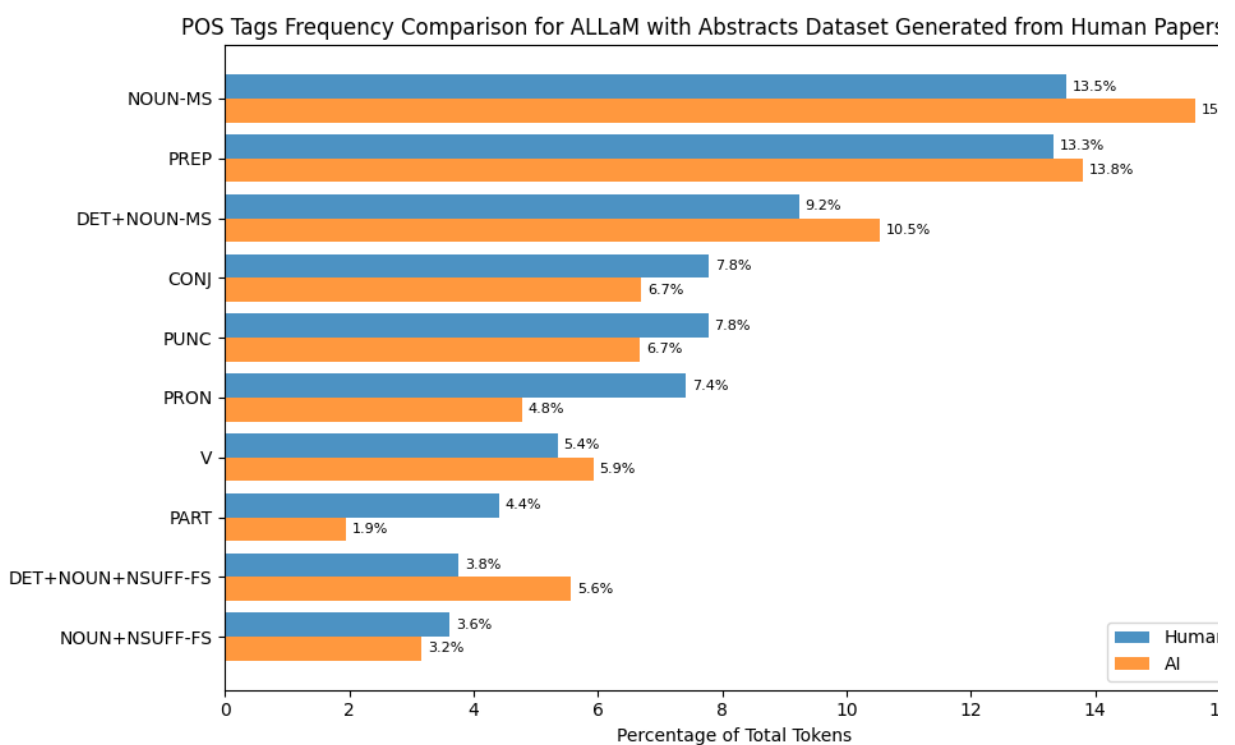
```
In [12]: analyze_frequencies(  
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",  
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",  
    title="NER Tags Frequency Comparison for ALLaM with Abstracts Dataset",  
    top_n=7,  
)
```



from title

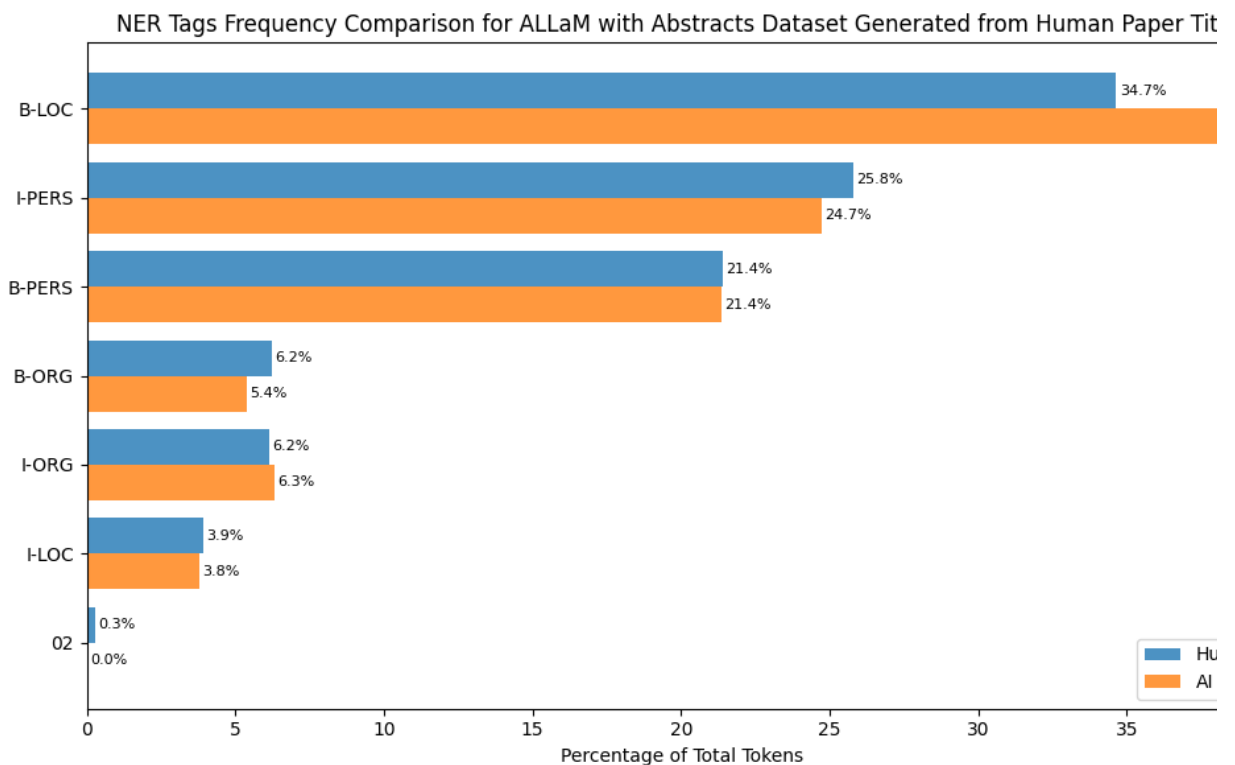
POS

```
In [13]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_dataset_generated_from_human_papers.json",
    title="POS Tags Frequency Comparison for ALLaM with Abstracts Dataset Generated from Human Papers",
    top_n=10,
)
```



NER

```
In [14]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_da
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data
    title="NER Tags Frequency Comparison for ALLaM with Abstracts Dataset
    top_n=7,
)
```

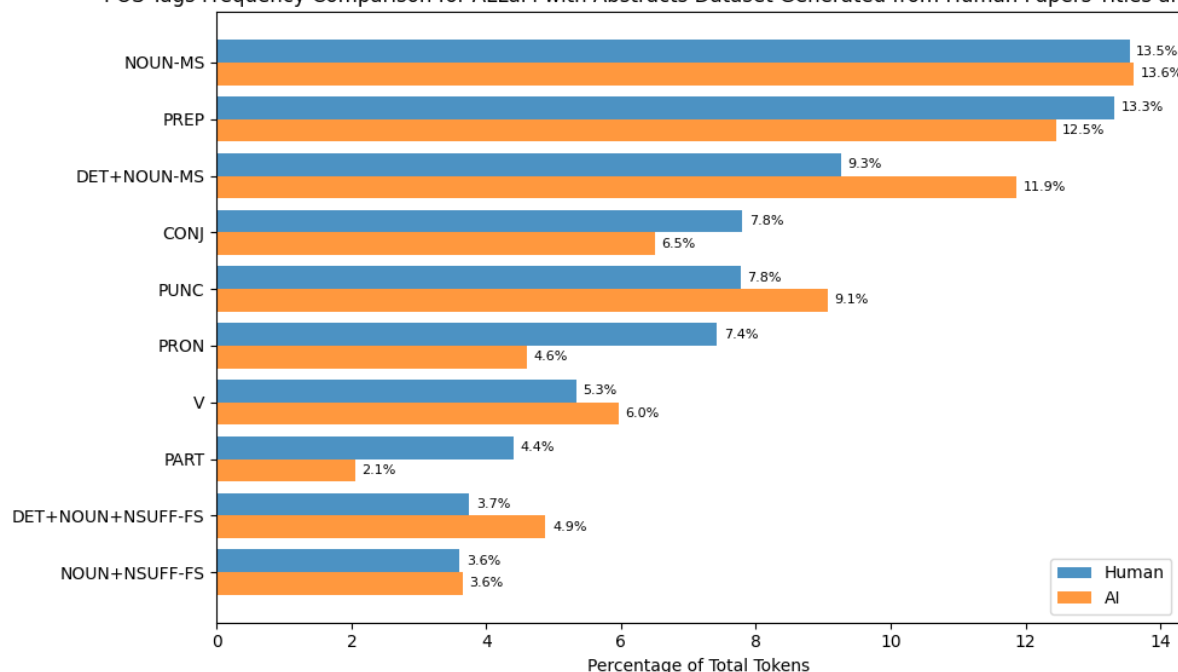


from title and content

POS

```
In [15]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_da
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data
    title="POS Tags Frequency Comparison for ALLaM with Abstracts Dataset
    top_n=10,
)
```

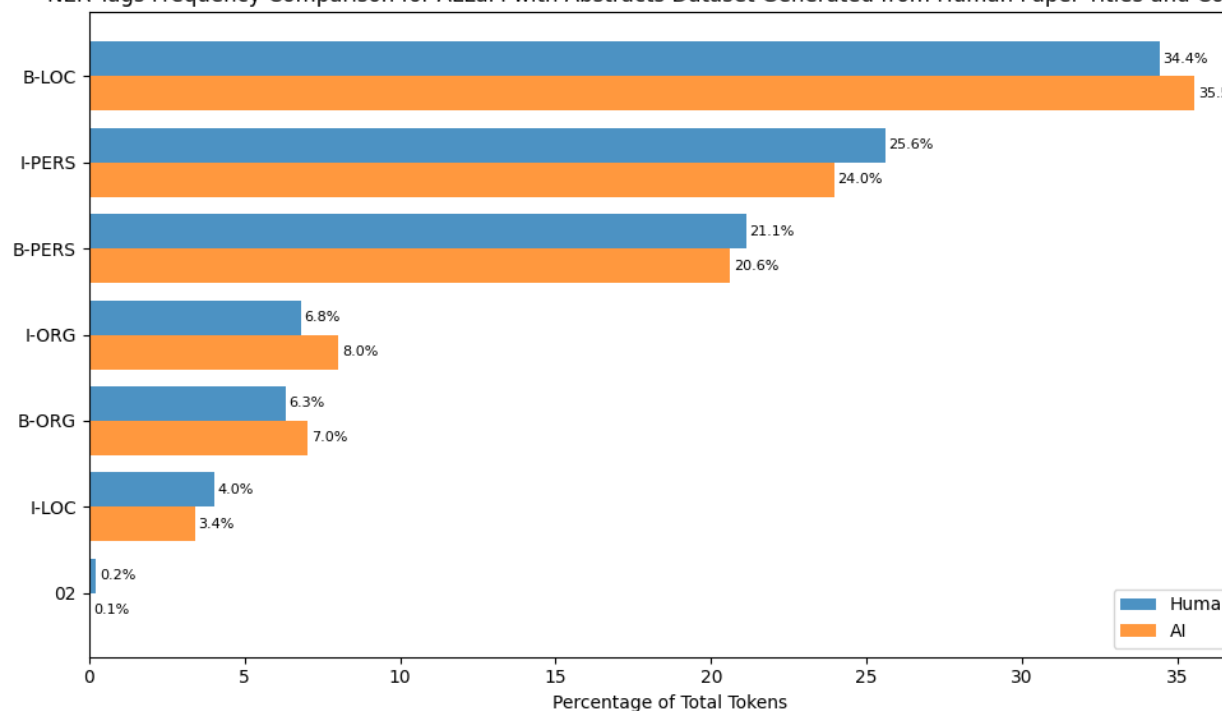
POS Tags Frequency Comparison for ALLaM with Abstracts Dataset Generated from Human Papers Titles and Co



NER

```
In [16]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for ALLaM with Abstracts Dataset",
    top_n=7,
)
```

NER Tags Frequency Comparison for ALLaM with Abstracts Dataset Generated from Human Paper Titles and Co

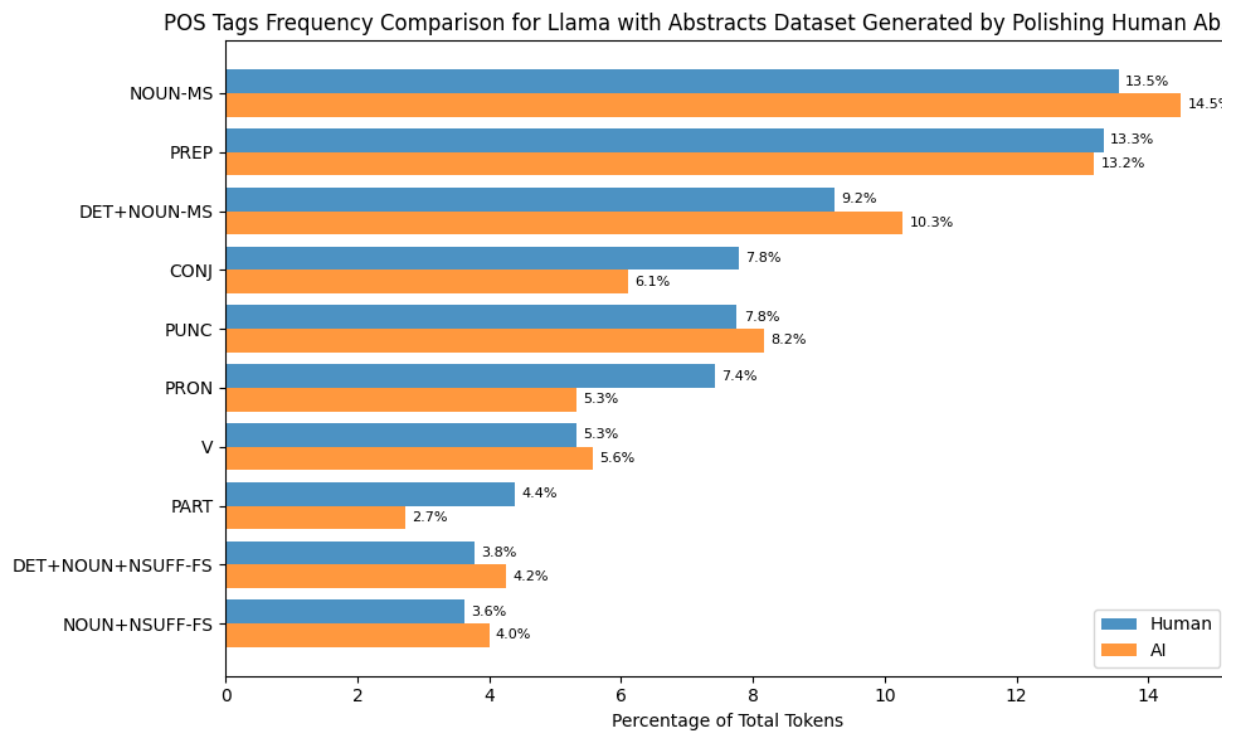


Llama

By Polishing

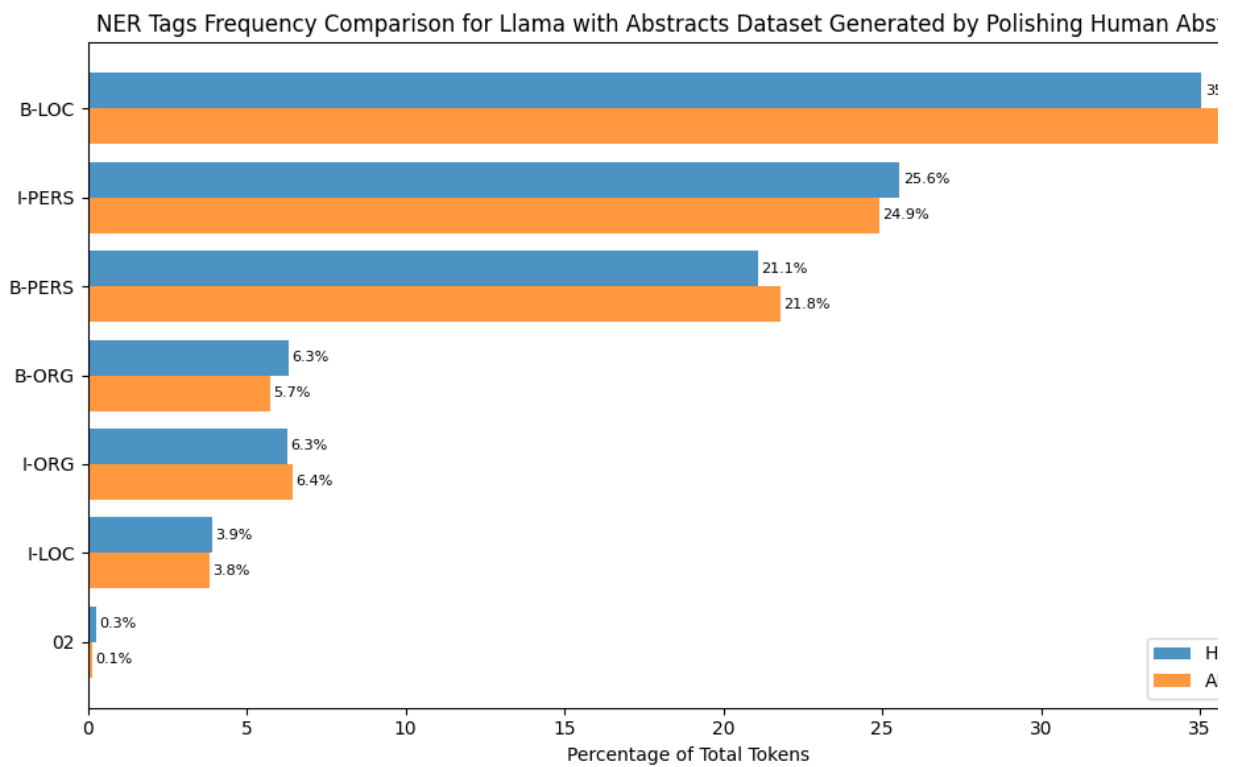
POS

```
In [17]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="POS Tags Frequency Comparison for Llama with Abstracts Dataset",
    top_n=10,
)
```



NER

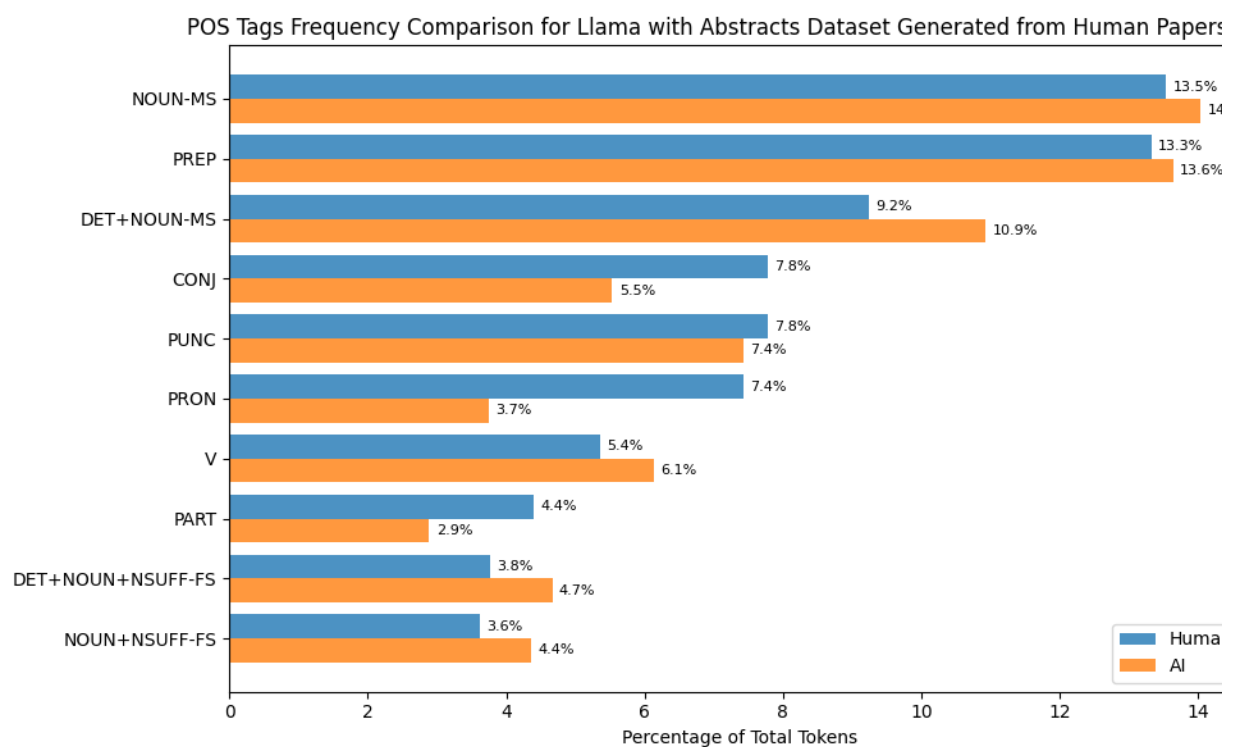
```
In [18]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for Llama with Abstracts Dataset",
    top_n=7,
)
```



From Title

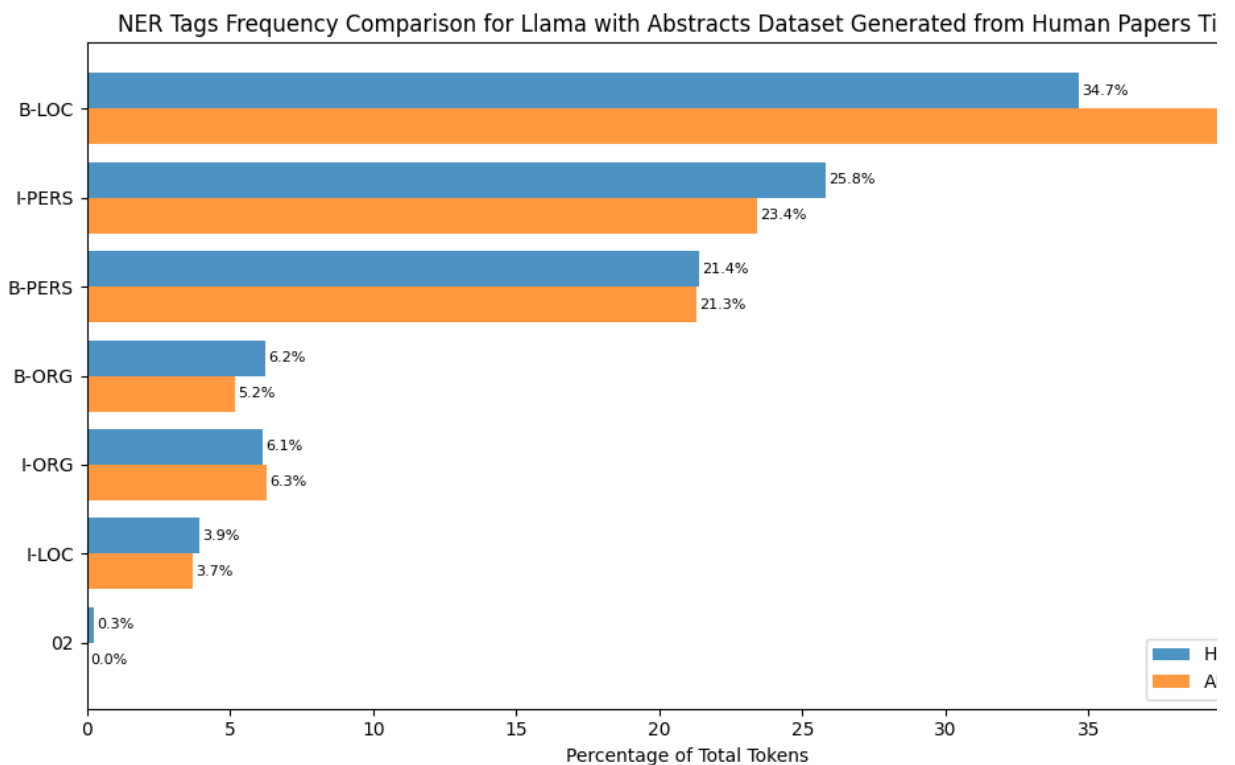
POS

```
In [19]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data/human_pos_tags.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data/ai_pos_tags.json",
    title="POS Tags Frequency Comparison for Llama with Abstracts Dataset",
    top_n=10,
)
```



NER

```
In [20]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_da
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data
    title="NER Tags Frequency Comparison for Llama with Abstracts Dataset
    top_n=7,
)
```

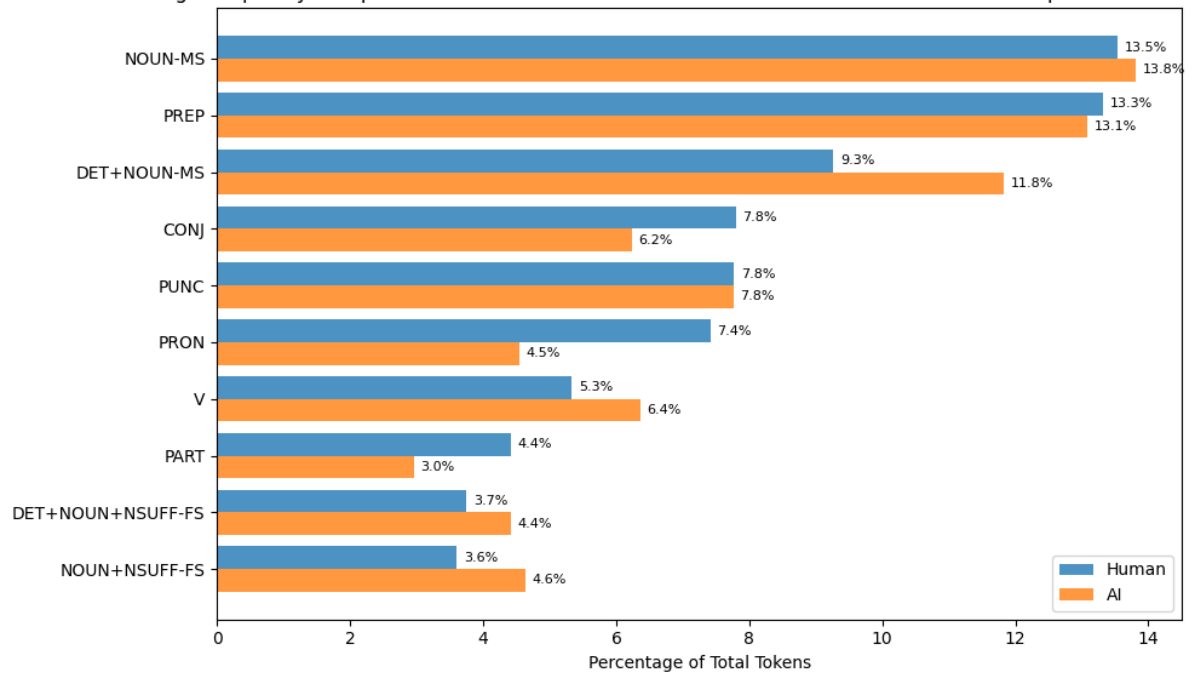


From title and content

POS

```
In [21]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_da
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data
    title="POS Tags Frequency Comparison for Llama with Abstracts Dataset
    top_n=10,
)
```

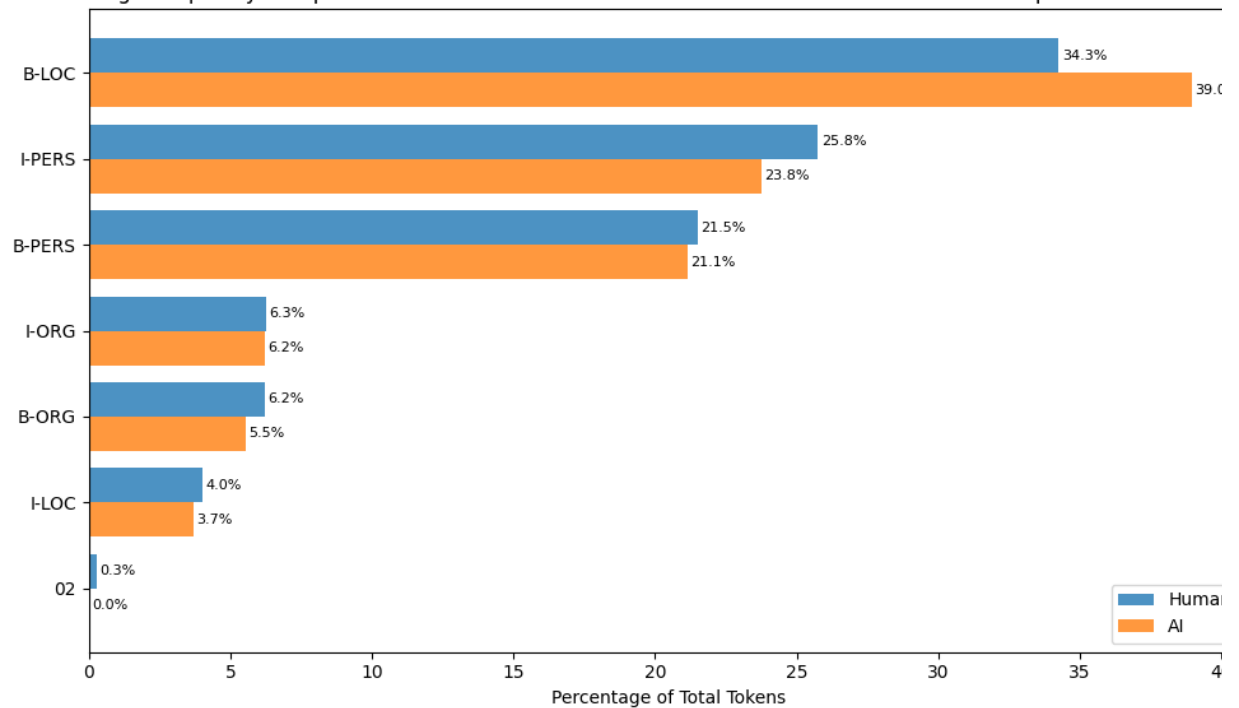
POS Tags Frequency Comparison for Llama with Abstracts Dataset Generated from Human Papers Titles and Co



NER

```
In [22]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for Llama with Abstracts Dataset",
    top_n=7,
)
```

NER Tags Frequency Comparison for Llama with Abstracts Dataset Generated from Human Papers Titles and Co

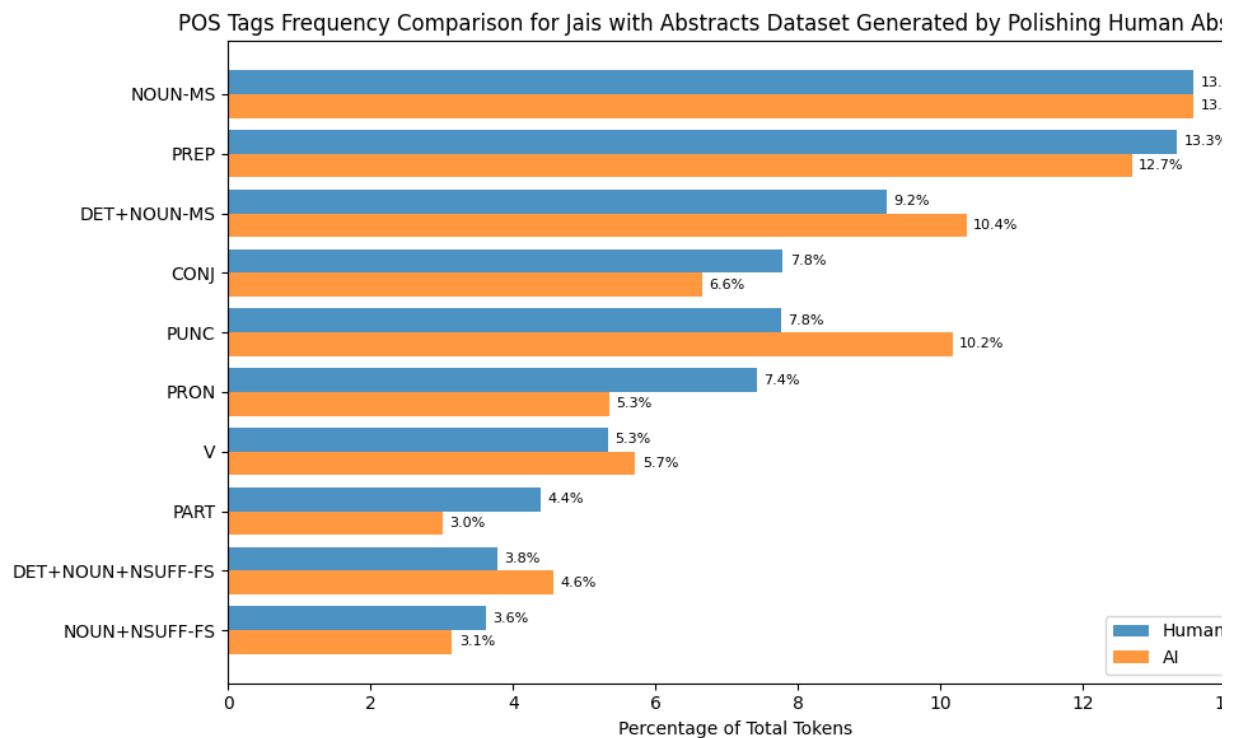


Jais

By Polishing

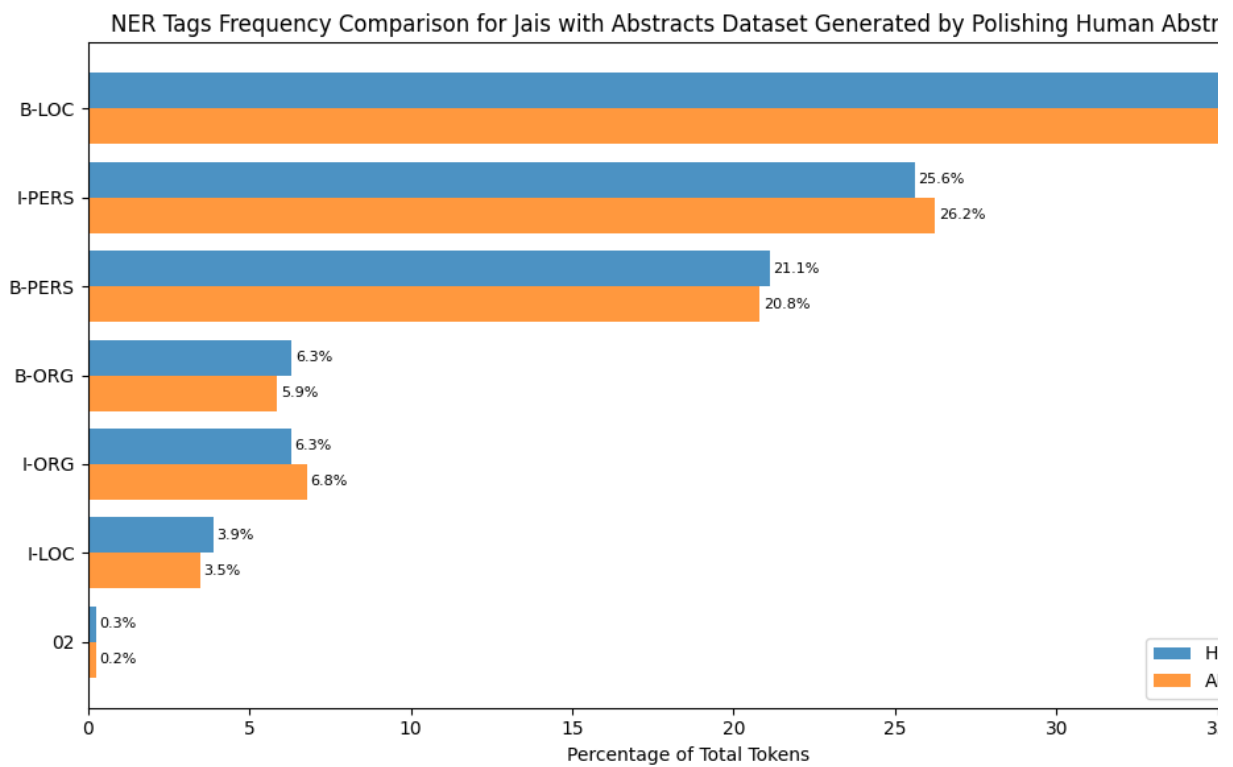
POS

```
In [23]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="POS Tags Frequency Comparison for Jais with Abstracts Dataset",
    top_n=10,
)
```



NER

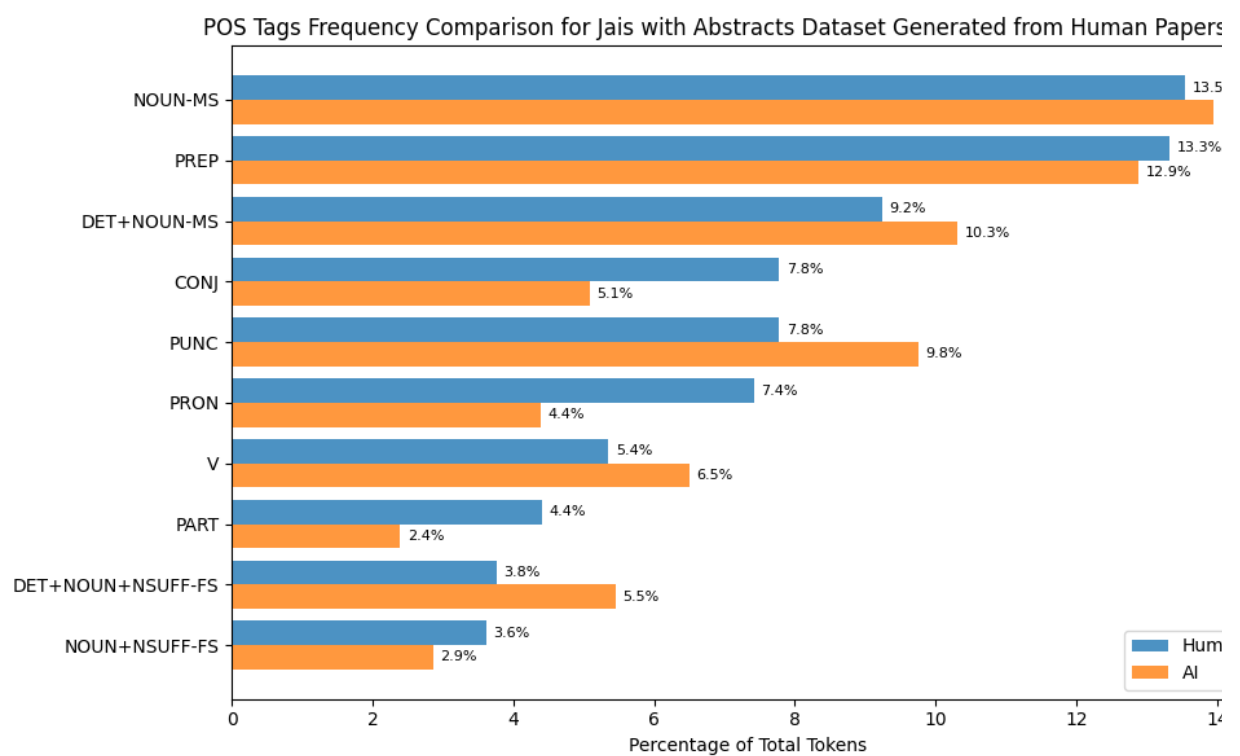
```
In [24]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for Jais with Abstracts Dataset",
    top_n=7,
)
```



From Title

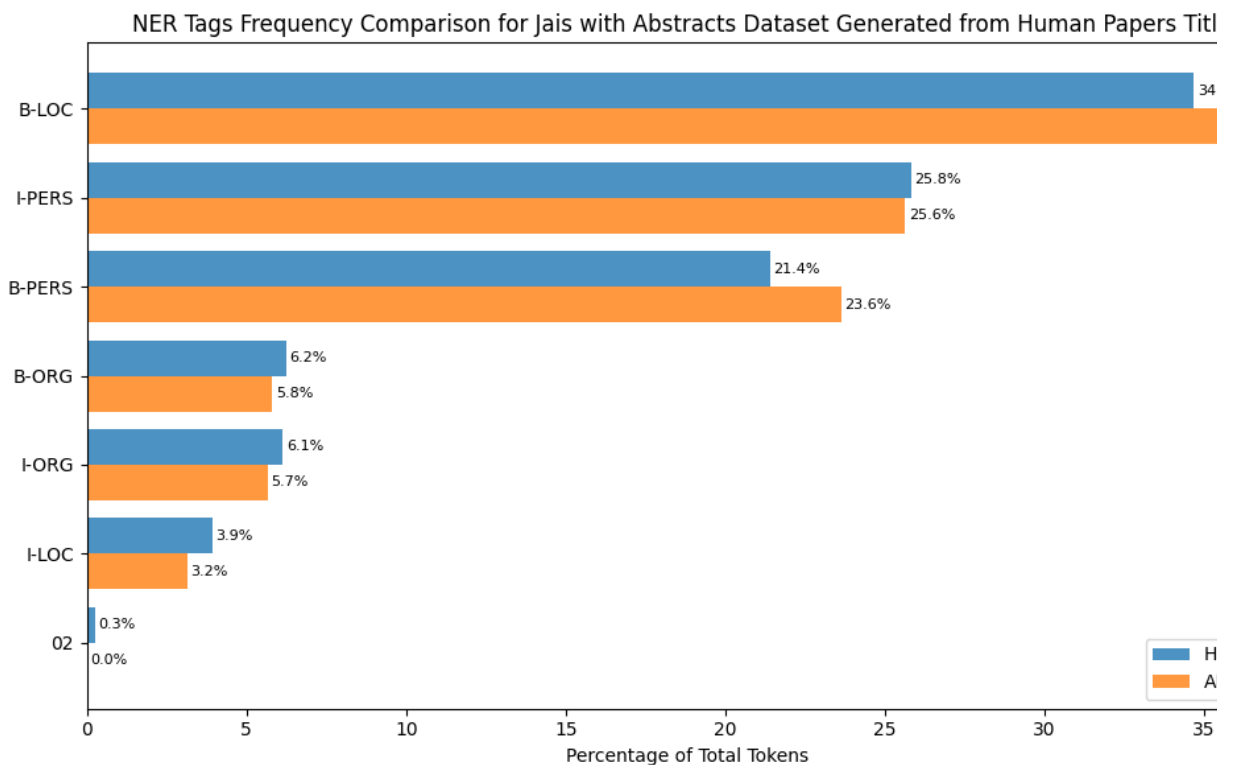
POS

```
In [25]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data/human_frequency.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data/ai_frequency.json",
    title="POS Tags Frequency Comparison for Jais with Abstracts Dataset",
    top_n=10,
)
```



NER

```
In [26]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for Jais with Abstracts Dataset",
    top_n=7,
)
```

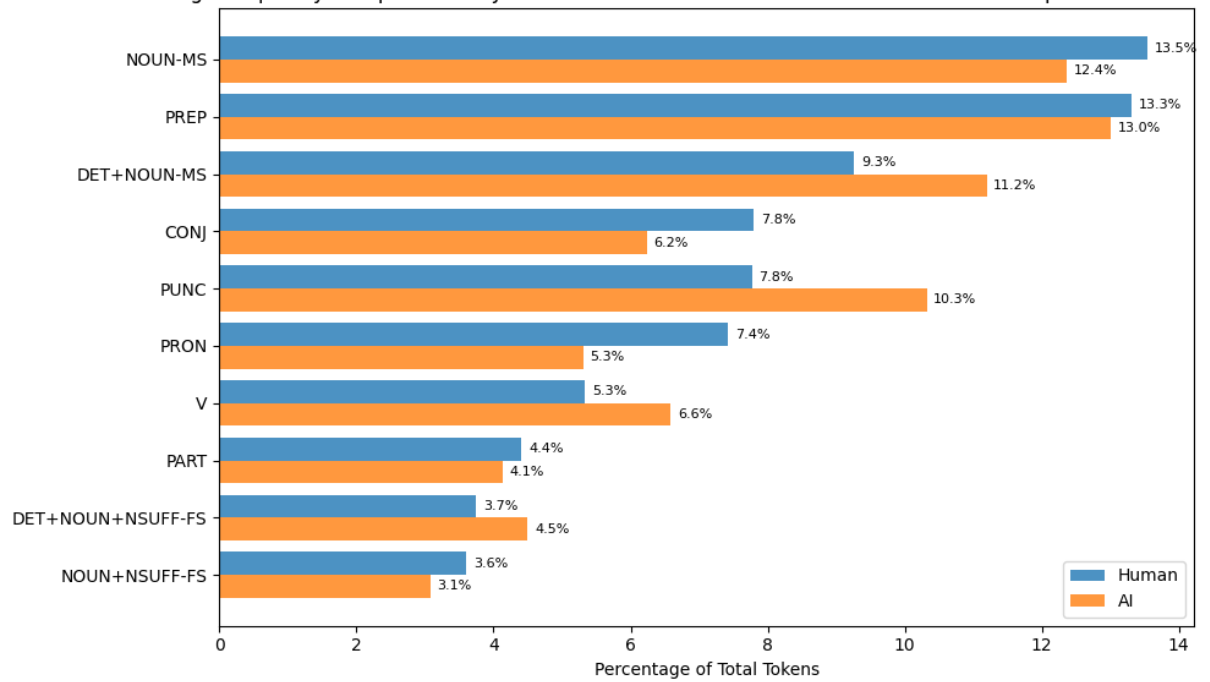


From Title and Content

POS

```
In [27]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="POS Tags Frequency Comparison for Jais with Abstracts Dataset",
    top_n=10,
)
```

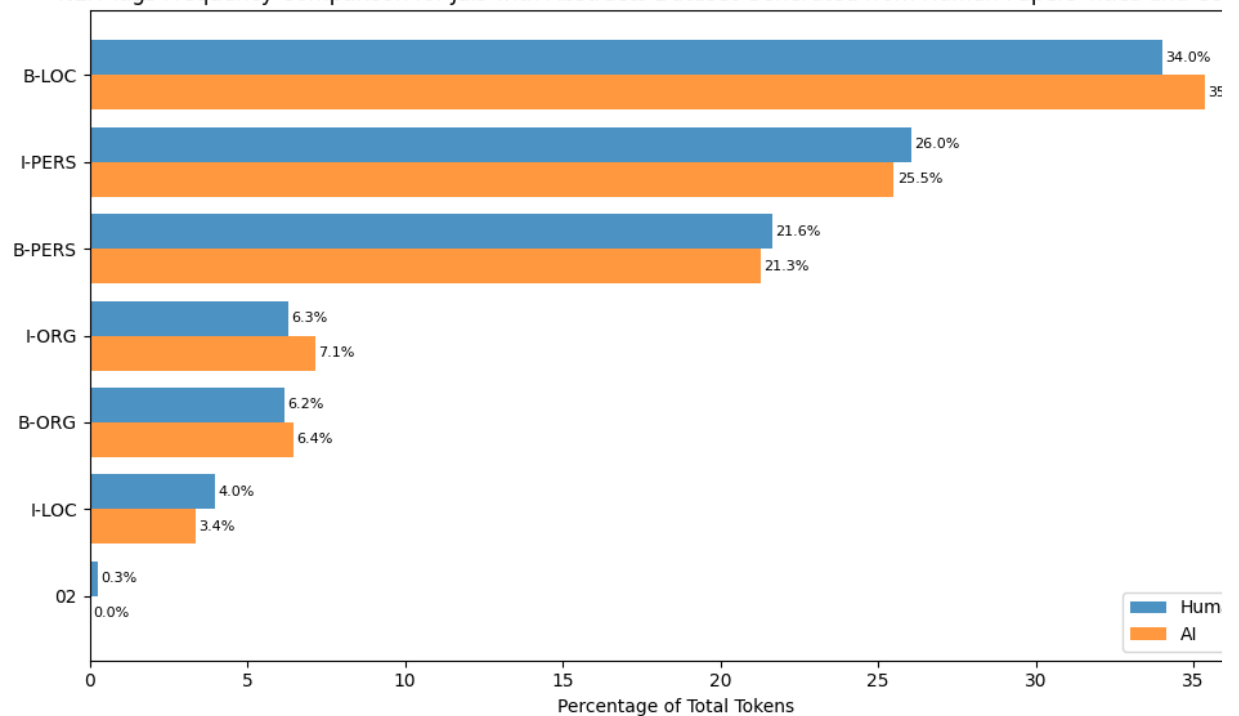
POS Tags Frequency Comparison for Jais with Abstracts Dataset Generated from Human Papers Titles and Co



NER

```
In [28]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for Jais with Abstracts Dataset",
    top_n=7,
)
```

NER Tags Frequency Comparison for Jais with Abstracts Dataset Generated from Human Papers Titles and Co

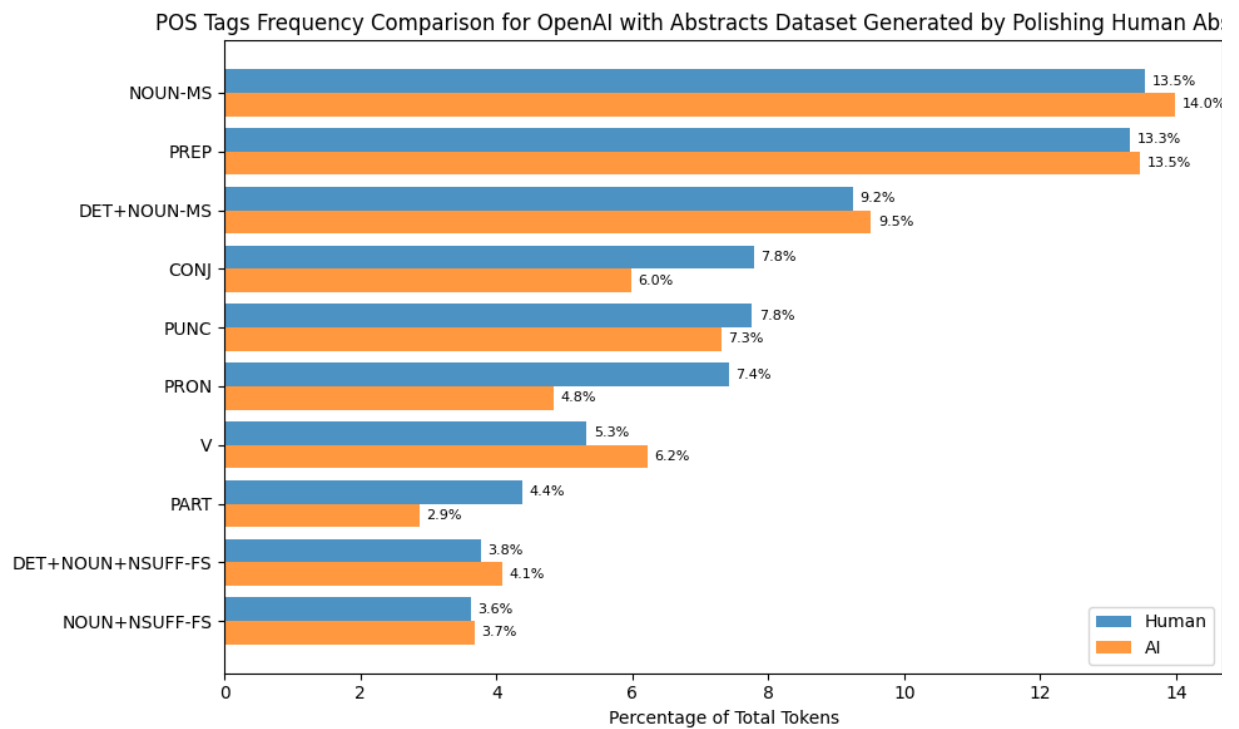


OpenAI

By Polishing

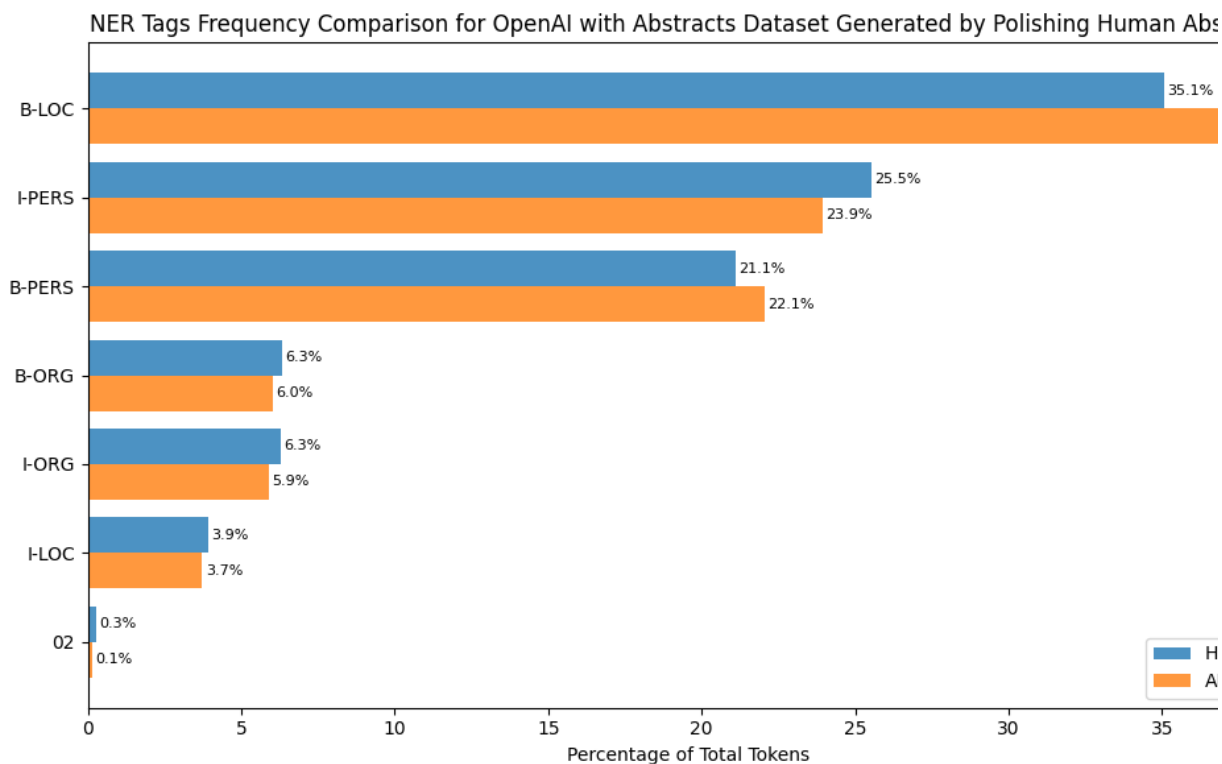
POS

```
In [29]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="POS Tags Frequency Comparison for OpenAI with Abstracts Dataset",
    top_n=10,
)
```



NER

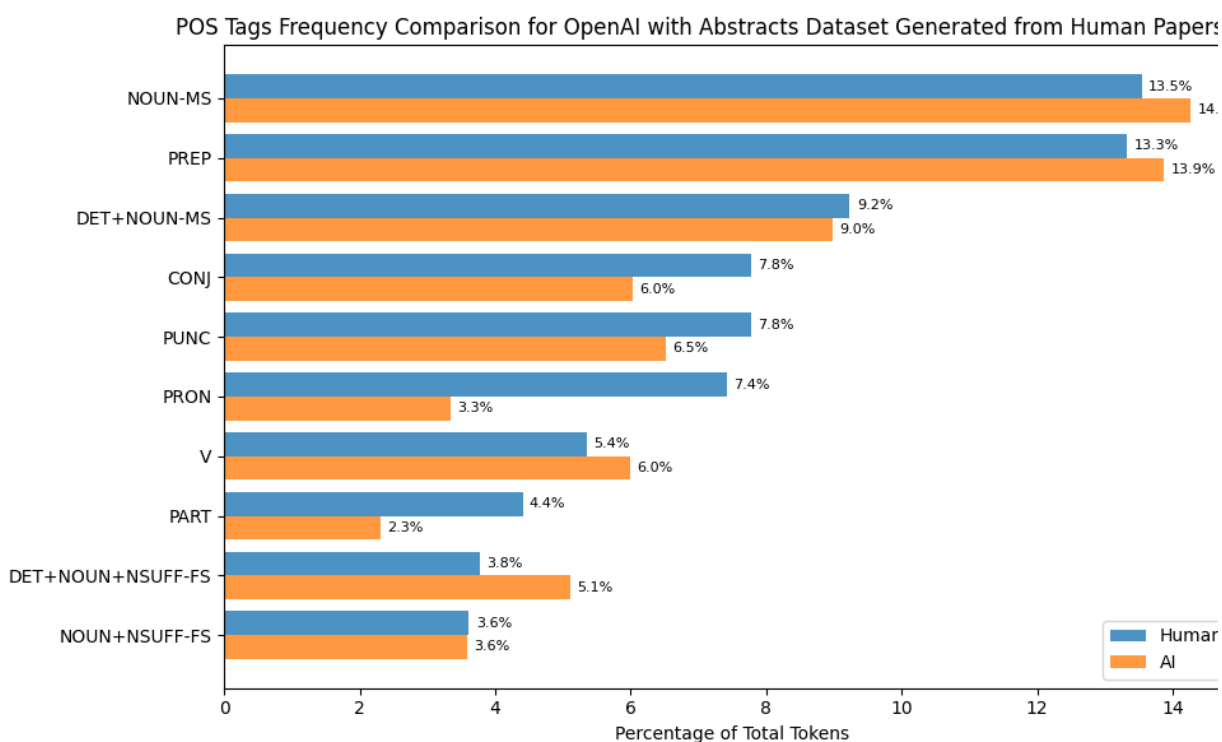
```
In [30]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for OpenAI with Abstracts Dataset",
    top_n=7,
)
```



From Title

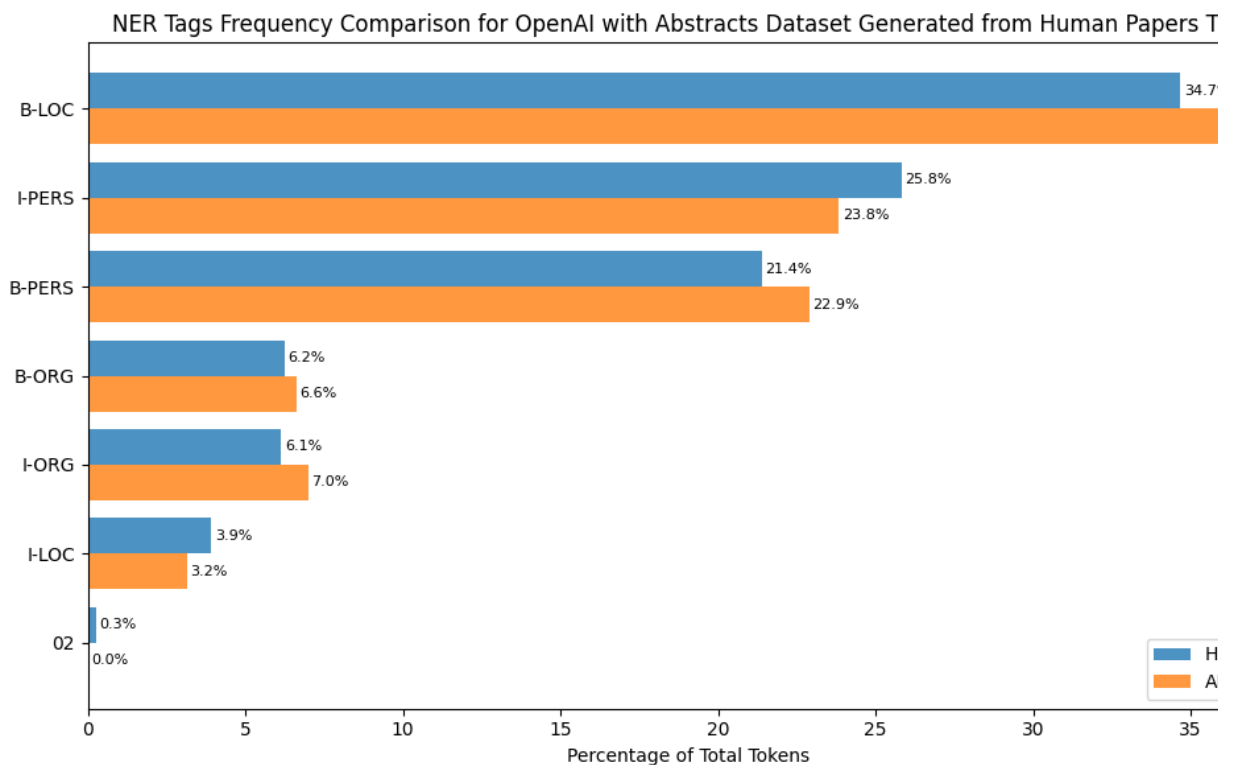
POS

```
In [31]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data/human_pos_tags.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data/ai_pos_tags.json",
    title="POS Tags Frequency Comparison for OpenAI with Abstracts Dataset",
    top_n=10,
)
```



NER

```
In [32]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_da
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data
    title="NER Tags Frequency Comparison for OpenAI with Abstracts Datas
    top_n=7,
)
```

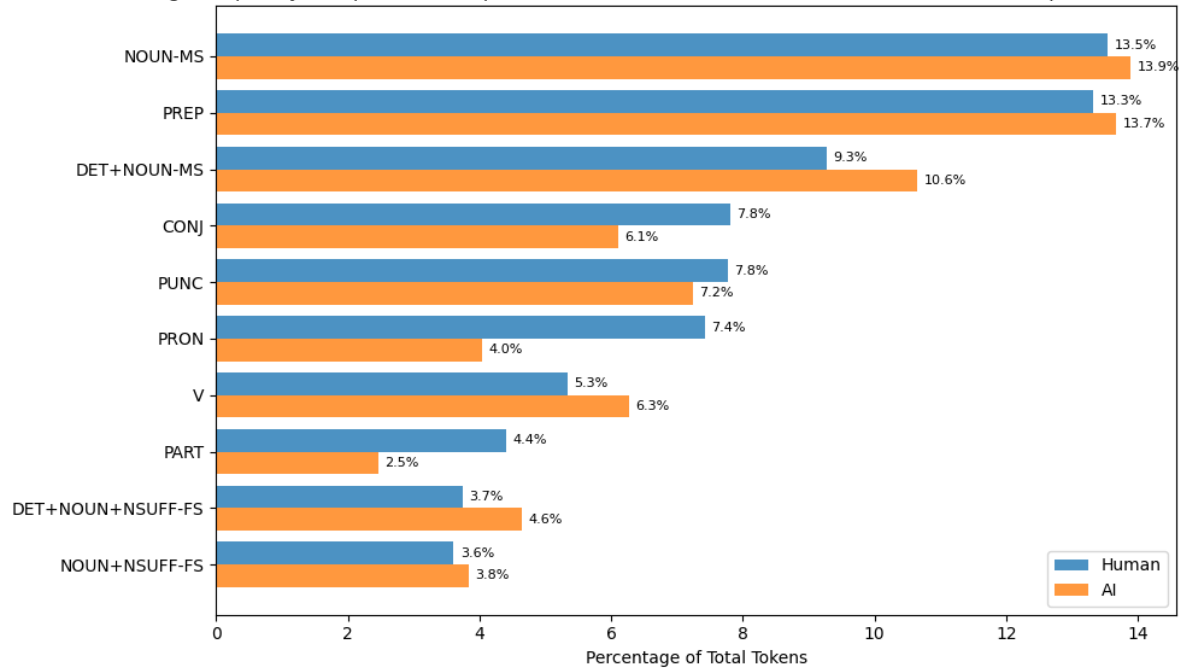


From Title and Content

POS

```
In [33]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_da
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data
    title="POS Tags Frequency Comparison for OpenAI with Abstracts Datas
    top_n=10,
)
```

POS Tags Frequency Comparison for OpenAI with Abstracts Dataset Generated from Human Papers Titles and Co



NER

```
In [34]: analyze_frequencies(
    human_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_human.json",
    ai_frequency_jsonpath="notebooks/stylometric_analysis/abstracts_data_ai.json",
    title="NER Tags Frequency Comparison for OpenAI with Abstracts Dataset",
    top_n=7,
)
```

NER Tags Frequency Comparison for OpenAI with Abstracts Dataset Generated from Human Papers Titles and Co

